

HVDC Power Supply Design

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Circuits like Geiger counters, insect zappers, Nixie tubes and sensors require high-voltage direct-current (HVDC) supplies. There are various types of HVDC supplies available in the market, including voltage doubler or quadrupler, flyback converter and boost converter. Some of these have low current-output capacity. But with right calculations using basic boost conversion formulae, we may achieve HVDC supplies capable of clean and high current capacity. Application notes supplied by component manufacturers provide many handy formulae compatible with their components that are derived from these basic formulae. Presented here is a

boost converter design using MC34063 DC-DC converter. The author's prototype is shown in Fig. 1.

Boost converter basics

In a boost converter (Fig. 2), energy is stored in the inductor (L1a) during the time the transistor (T1a) is 'on' (ton). When the transistor is turned off (toff), the energy is transferred in series with input Vin to the output filter capacitor (Cout) and load (RL). This configuration allows setting of the output voltage to any value greater than the input.

The output voltage can be calculated as follows:

$$V_{out} = V_{in}(\text{ton}/\text{toff}) + V_{in}, \text{ or}$$

$$V_{out} = V_{in}((\text{ton}/\text{toff}) + 1)$$

Circuit and working

Circuit diagram of the boost converter using MC34063 DC-DC converter is shown in Fig. 3.

MC34063 is a monolithic control circuit containing all the active functions required for switching DC-to-DC converters. It represents significant advancements in ease of use with highly efficient,

yet simple switching regulators. The use of switching regulator is becoming more pronounced than linear regulators because of the size and power-efficiency requirements of new equipment designs. Switching regulators increase application flexibility while reducing the cost.

MC34063 was designed for buck, boost and voltage-inverter converter applications. It includes temperature-compensated reference voltage, oscillator, active peak-current limit, output switch and output-voltage-sense comparator. All these functions are contained in an 8-pin DIP or SOIC package.

Internal diagram of MC34063 as per the datasheet given by Texas Instruments is shown in Fig. 4.

Its pin 5 (comparator inverting input) senses and sets the output voltage to a stable value for calculating feedback resistor values as shown in Fig. 5.

$$V_{out} = 1.25((R2a/R1a) + 1)$$

The internal voltage regulator produces 1.25 volts for the internal comparator, so the external voltage divider comprising R1a and R2a should be arranged such that it gives exactly 1.25 volts when the desired output voltage is reached. For example, if you need output voltage of around 501 volts, the voltage-divider resistor values must be R2a = 2.4 mega-ohms and R1a = 6 kilo-ohms, respectively.

As shown in the block diagram, the comparator output triggers and disables the SR latch. The oscillator driven by the timing capacitor at pin 3 is composed of a current source and sink elements, which charge and discharge the external timing capacitor between upper and lower preset thresholds. Typically, charge and discharge currents are 35mA and 200mA, respectively, yielding approximately a 6:1 ratio. Thus, the ramp-up period is six times longer than the ramp-down period. The upper threshold is equal to internal reference voltage of 1.25V, and the lower threshold is approximately 0.75 V.

The oscillator runs continuously at a rate controlled by the timing capacitor value. It also senses peak current by sensing the voltage generated by the inductor current across a small-value, higher-wattage sensing resistor connected to pin 7. In this circuit (Fig. 3), 1.5-ohm, 2W resistor R6 is the sensing resistor.

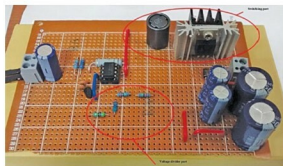


Fig. 1: Author's prototype

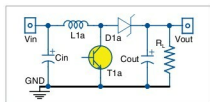


Fig. 2: Boost switching regulator circuit

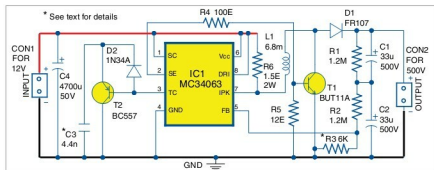


Fig. 3: Circuit diagram for HVDC power supply